

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Analytical Problem Solving

Course Code:	ITA0532	Course Title:	Computer Vision
CO Assessed:	CO2 – Imitation (BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	10% of CIA	Total Marks:	100
CO1 Statement:	<i>Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)</i>		
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Section / Batch:	ECE	Date:	09-08-2026

16. High-Pass Filtering

An image requires sharpening.

(a) Interpret need for high-pass filtering.

(b) Justify filter selection.

(c) Apply mask:

(d) Compute center value.

(e) Interpret edge enhancement.

ANS :

High-Pass Filtering

An image requires sharpening.

(a) Interpret need for high-pass filtering

High-pass filtering is used for image sharpening. It emphasizes rapid changes in intensity, particularly edges and fine details, while reducing slowly varying background information.

(b) Justify filter selection

A high-pass filter is appropriate because sharpening requires enhancement of high-frequency components. A common sharpening mask is:

The positive center coefficient strengthens the original pixel, while the negative neighboring coefficients emphasize differences between the center and its surroundings.

(c) Apply mask

For a neighborhood

the filtered value is

$$(-a-b-c-d-g-h-i)+9e$$

Or

$$9e-(a+b+c+d+f+g+h+i).$$

(d) Compute center value

If the center pixel is , its output is:

$$9e-(a+b+c+d+f+g+h+i)$$

For example, if the eight neighboring pixels sum to 400 and the center is 60:

$$9(60)-400=540-400=140$$

(e) Interpret edge enhancement

Edges have large intensity differences between neighboring pixels, so the high-pass mask produces a strong response there. Thus, edges become sharper and image details are enhanced.

17. Frequency Domain High-Pass

Given frequency components:

- (a) Interpret high-frequency importance.**
- (b) Justify high-pass filtering.**
- (c) Apply filtering (remove low frequencies).**
- (d) Compute output.**
- (e) Interpret result.**

ANS:

Frequency Domain High-Pass

Given frequency components:

(a) Interpret high-frequency importance

High-frequency components represent **rapid intensity variations**, such as edges, boundaries, textures, and fine details. They are therefore important for image sharpness.

(b) Justify high-pass filtering

A high-pass filter is used when the objective is to **retain high-frequency information while suppressing low-frequency information** such as smooth backgrounds and gradual illumination changes.

(c) Apply filtering — remove low frequencies

Suppose the frequency-domain components are:

$$[10, 20, 50, 80, 120]$$

and the first three are considered low-frequency components. A high-pass filter removes them:

$$[10, 20, 50, 80, 120] \rightarrow [0, 0, 0, 80, 120]$$

(d) Compute output

The filtered frequency-domain output is therefore:

$$[0, 0, 0, 80, 120]$$

If an inverse Fourier transform is applied, these retained high-frequency components produce an image containing mainly **edges and fine details**.

(e) Interpret result

The output has reduced smooth/background information and enhanced rapid intensity changes. Therefore, the image appears **sharper and more edge-focused**, although excessive high-pass filtering can also amplify noise.

18. Adaptive Thresholding

Pixel values: [60, 80, 120, 140]

(a) Interpret segmentation challenge.

(b) Justify adaptive thresholding.

(c) Apply local threshold (mean = 100).

(d) Compute binary output.

(e) Interpret result.

ANS:

Adaptive Thresholding

Pixel values:

[60,80,120,140]

Local mean:

$$T=100$$

(a) Interpret segmentation challenge

A single global threshold may not work well when an image has **uneven illumination or varying background intensity**. Different regions may require different threshold values.

(b) Justify adaptive thresholding

Adaptive thresholding determines the threshold from the **local neighborhood**. This makes segmentation more effective when brightness varies across the image.

(c) Apply local threshold

Using the local mean:

$$T=100$$

Use the rule:

$$B(x) = \begin{cases} 1, & I(x) > 100 \\ 0, & I(x) \leq 100 \end{cases}$$

(d) Compute binary output

Pixel Threshold = 100 Binary output

60	(60 ≤ 100)	0
80	(80 ≤ 100)	0
120	(120 > 100)	1
140	(140 > 100)	1

Therefore: [0,0,1,1]

(e) Interpret result

Pixels below the local mean are classified as **background (0)**, while pixels above the mean are classified as **foreground (1)**. Thus, the values 120 and 140 are segmented as the foreground.

Code of Conduct

/ ALLU THANMAI (192412616) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: thanmai A

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Criteria	Marks
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	Problem Understanding	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	Method Selection	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	Solution Process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	Accuracy of Calculation	
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	Interpretation of Results	